



Google Step Up Challenge: How to Guide

Your Mission: Power Google’s Gemini Pro Breakthrough

Help Google launch a multi- market digital marketing campaign to reach and excite university students about Gemini Pro.

Budget: \$10M USD

Length: 8 weeks

Markets: 3

Identify what has historically driven success with this audience and transform these insights into a powerful, data-driven campaign strategy:

- Reach the student target audience efficiently
- Drive Consideration
- Boost sign-ups of Gemini Pro

The Data Dictionary

To solve this challenge, you have been given three specific datasets. Below is the exact breakdown of the columns available in each file.

A. Historic Campaign Data

Filename: historic_campaign_data

This file contains the actual weekly performance metrics for past campaigns. Use this to determine efficiency (e.g., Cost Per Acquisition, cpm) by market, channel and time period

Column Name	Description
Week_Start	The date the reporting week began.
Week_Date	The specific date of the record.
Campaign_Name	The name of the campaign.
Market	The country where the campaign ran (e.g., UK, SA).

Channel	The platform used (e.g., YouTube, Social).
Spend_USD	The budget spent in US Dollars
Impressions	Total number of times the ad was shown.
Reach	Total number of <i>unique</i> people who saw the ad.
Conversions	The number of people who signed up for Gemini Pro.

B. Brand Lift Study

Filename: brand_lift_study

This file contains survey data comparing Gemini Pro Consideration rates for users who were exposed to our ads vs those who were not. Use this to calculate consideration lift, statistical significance and cost per lift for each historic campaign, market and channel

Column Name	Description
Campaign_Name	Matches the Campaign Data.
Market	The country where the study ran.
Channel	The platform being measured.
Control_Responses	The total number of people surveyed who did not see the ad (Sample Size).
Control_Consideration	The number of people in the Control group who would consider Gemini Pro
Exposed_Responses	The total number of people surveyed who did see the ad (Sample Size).
Exposed_Consideration	The number of people in the Exposed group who would consider Gemini Pro
Control_Rate	The % of the Control group who would consider Gemini Pro

Exposed_Rate	The % of the Exposed group who would consider Gemini Pro
Absolute_Lift	The percentage increase in consideration driven by the ad. Calculated as (Exposed_Rate - Control_Rate)

C. Creative Performance Report

Filename: *Creative_performance_report*

This file contains 'In-lab' creative testing results for *new* ad concepts. Use this to decide which creatives (ads) to use in the upcoming campaign. Different creatives may drive different KPIs or suit different markets, channels or age groups better. Please note all lift results should be considered statistically significant.

Column Name	Description
Report_Date	Date the test result was reported.
Creative_Name	The theme of the new ad (e.g., "Future is Now").
Market	The country where respondents resided
Channel	The platform tested.
Age_Group	The age range of respondents (e.g., "18-24").
Awareness_Lift	Impact on Brand Awareness. (absolute lift)
Consideration_Lift	Impact on Consideration. (absolute lift)
Purchase_Intent_Lift	Impact on Purchase Intent. (absolute lift)



Data Science Route

Your Goal: Decode what drove historic impact.

Focus your analysis on the **Historic Campaign Data** and **Brand Lift Study** outcomes

Output: Short slide deck with 3-5 key insights and supporting data.

- What trends exist in the campaign data?
 - Which markets and channels performed best? (ie. had the most sales and best efficiency?)
 - What is the relative cost to target by market/channel?
- Which things worked best for brand lift?
 - What campaigns were most effective (drove the most relative lift)
 - Which were statistically significant?
 - BONUS: Which was most efficient (requires spend from campaign data)
- What creatives (adverts) resonated best with our target audience?
 - Does this differ by market?

Processing the data: How to guide

Below are a few key steps to help you unlock more from the datasets and provide deeper insights to guide the next campaign plan.

Step 1: Calculate Statistical Significance

Before putting together any insights, you need to establish which brand lift studies are statistically significant ie. the lift in KPI is a result of exposure to our advertising and not just random chance. You should only consider the lift from statistically significant studies as usable for analysis. Given the volume of brand lift studies in the data set, you need a robust, easily repeatable way to check if the difference between your **Control** and **Exposed** groups is real (ie. statistically significant) or just random chance. You can use Excel or Python to calculate this efficiently.

Excel Method (Z-Test)

You can build a significance calculator using the following steps. Assume your data starts in row 2:

1. **Calculate Pooled Proportion ($\hat{p}$):**
Formula: $= (\text{Control_Consideration} + \text{Exposed_Consideration}) / (\text{Control_Responses} + \text{Exposed_Responses})$
2. **Calculate Standard Error (SE):**
Formula: $= \text{SQRT}(\hat{p} * (1 - \hat{p}) * ((1 / \text{Control_Responses}) + (1 / \text{Exposed_Responses})))$
(Note: Use the cell reference for $\hat{p}$ calculated in step 1)
3. **Calculate Z-Score (Z):**
Formula: $= (\text{Exposed_Rate} - \text{Control_Rate}) / SE$
4. **Calculate P-Value:**
Formula: $= 2 * (1 - \text{NORM.S.DIST}(\text{ABS}(Z), \text{TRUE}))$
5. **Check Significance:**
Formula: $= \text{IF}(\text{P_Value} < 0.05, \text{"Significant"}, \text{"Not Significant"})$

Python method (Statsmodels Package)

If you prefer coding, the statsmodels library is the industry standard for this calculation.

```
from statsmodels.stats.proportion import proportions_ztest

# Define your counts [Control Successes, Exposed Successes]
counts = [control_consideration, exposed_consideration]

# Define your sample sizes [Control Total, Exposed Total]
sample_size = [control_responses, exposed_responses]

# Calculate the statistic and p-value
stat, pval = proportions_ztest(counts, sample_size)

# Check results
if pval < 0.05:
    print("Statistically Significant")
else:
    print("Not Significant")
```

Step 2: Join the Brand lift outcomes to the historic campaign data

To find the "Cost per Lifted User," you need to combine the **Spend** from the *Historic Campaign Data* with the stat sig **Absolute Lift** from the *Brand Lift Study* data.

- **Identify common column names:** You must match the rows in each data set based on the three columns common to both data sets:
 1. Campaign_Name
 2. Market
 3. Channel

Excel Method:

1. Create a "Join key" in both datasets that combines these three columns (e.g., =C2&D2&E2 to create BackToSchool_24UKYouTube).
 2. Next use a VLOOKUP to match your statistically significant brand lift results to your historic spend data eg. =vlookup(J2,brand_lift_study!A2:K50,11,false)
- **Important:** Ensure your join key is on the left of any columns you wish to bring in to the historic spend dataset before attempting vlookup

Python method:

```
# Left join on three columns  
result = historic_campaign_data.merge(brand_lift_study, on=['col_1',  
'col_2', 'col_3'], how='left')  

```

Step 3: Calculate Cost Per Lifted User (CPLU)

Once you have absolute lift of the statistically significant brand lift studies in your historic campaign data you can calculate the efficiency of the campaigns for lifting consideration. This is the most critical metric for the challenge. It combines cost, reach, and impact.

The Formula:

1. Calculate Absolute Lift %

$$Exposed_Rate - Control_Rate = Absolute_Lift_$$

2. Calculate Volume of Lifted Users

$$Absolute_Lift_$$

(Note: Use the Reach column from the Historic Campaign Data)

3. Calculate Cost Per Lifted User

$$Spend_GBP \div Number_of_Lifted_Users = CPLU$$

(Lower is better—it means you spent less to influence each person).



The Data Strategist Route

Your Goal: Design the ultimate student campaign.

Using the **high level results provided** here (Data scientists.. NO peaking!) or the outputs of your own analysis, if you also followed the data scientist route, generate a series of recommendations for the upcoming campaign to maximise the impact of the student campaign within the budget.

Use Gemini to add insights and understand:

- Size of target audience by market
- Media consumption behaviours of your audience
- Approaches to budget allocation

Output: The recommendations should be in a short slide deck and need to cover the following areas:

- **Market selection:** Which markets should this campaign be run in to maximise overall impact?
- **Media execution:** Which channels, placement and tactics should we use for the campaign
- **Budget allocation:** Consider target audience size and historic cost to target to determine budget allocation between channels/markets
- **Creative feature focus:** Which creatives should be selected for the campaign based on the creative testing outcomes?